

ASSIGNMENT :NO 1

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Section : B
Electrical (AI)

```
#include <iostream> // Library for input/output
using namespace std; // Allows using cout, cin without std::
// Built in Function to convert grade (A, B, etc.) into grade points
double getGradePoint(char grade) { // Function starts
if (grade == 'A') return 4.0; // If grade is A, return 4.0
else if (grade == 'B') return 3.0; // If grade is B, return 3.0
else if (grade == 'C') return 2.0; // If grade is C, return 2.0
else if (grade == 'D') return 1.0; // If grade is D, return 1.0
else return 0.0; // For F or invalid grade, return 0
}
int main() { // Main function starts
// i. Declare variables
double voltage, current, resistance; // Declare three double variables

// ii. Display message
```

```
cout << "Welcome to OOPs in C++!" << endl; // Print welcome message
```

// iii. Read age

```
int age; // Declare integer variable for age
cout << "Enter your age: "; // Ask user to enter age
cin >> age; // Take input from user and store in age
```

// iv. Temperature check

```
double temperature; // Declare temperature variable
cout << "Enter temperature: "; // Ask for temperature
cin >> temperature; // Input temperature
if (temperature > 100) // Check if temperature is greater than 100
    cout << "Overheating detected!\n"; // Print warning message
```

// v. Area of circle

```
double r, area; // Declare radius and area
cout << "Enter radius: "; // Ask for radius
cin >> r; // Input radius
area = 3.14159 * r * r; // Calculate area using formula  $\pi r^2$ 
cout << "Area of circle: " << area << endl; // Display area
```

// vi. Print x, y, z

```
int x, y, z; // Declare three integers
cout << "Enter x, y, z: "; // Ask user
cin >> x >> y >> z; // Input values
```

```
cout << x << ", " << y << ", " << z << endl; // Print values separated by comma
```

// vii. Sum of voltage and current

```
cout << "Enter voltage and current: "; // Ask user
cin >> voltage >> current; // Input voltage and current
double total = voltage + current; // Calculate sum
cout << "Total: " << total << endl; // Display result
```

// viii. Division check

```
double numerator, denominator; // Declare variables
cout << "Enter numerator and denominator: "; // Ask user
cin >> numerator >> denominator; // Input values
if (denominator == 0) // Check if denominator is zero
    cout << "Error: Division by zero" << endl; // Display error
```

// ix. Increment count

```
int count; // Declare count
cout << "Enter count: "; // Ask user
cin >> count; // Input value
cout << "After increment: " << ++count << endl; // Increment and print
```

// x. Read three floats

```
float a, b, c; // Declare three float variables
cout << "Enter three numbers: "; // Ask user
cin >> a >> b >> c; // Input values
```

// xi. Series resistance

```
double r1, r2, r3;                // Declare resistances
cout << "Enter three resistances (series): "; // Ask user
cin >> r1 >> r2 >> r3;           // Input values
cout << "Total Series Resistance: " << (r1 + r2 + r3) << endl; // Formula  $R = r_1 + r_2 + r_3$ 
```

// xii. Parallel resistance

```
cout << "Enter three resistances (parallel): "; // Ask user
cin >> r1 >> r2 >> r3;           // Input values
cout << "Total Parallel Resistance: " << 1 / (1/r1 + 1/r2 + 1/r3) << endl; // Parallel formula
```

// xiii. Voltage division

```
double V, R1, R2;                // Declare variables
cout << "Enter V, R1, R2: ";      // Ask user
cin >> V >> R1 >> R2;           // Input values
cout << "Voltage across R2: " << V * (R2 / (R1 + R2)) << endl; // Voltage division formula
```

// xiv. Current division

```
double I;                        // Declare current
cout << "Enter total current I, R1, R2: "; // Ask user
cin >> I >> R1 >> R2;           // Input values
cout << "Current through R2: " << I * (R1 / (R1 + R2)) << endl; // Current division formula
```

// xv. GPA calculation using alphabet grades

```
char g1, g2, g3;                // Declare grade variables (A, B, etc.)
```

```
double ch1, ch2, ch3;                // Declare credit hours
cout << "Enter grades (A/B/C/D/F) and credit hours:\n"; // Ask user
cout << "Subject 1: ";                // Prompt for subject 1
cin >> g1 >> ch1;                     // Input grade and credit hours
cout << "Subject 2: ";                // Prompt for subject 2
cin >> g2 >> ch2;                     // Input grade and credit hours
cout << "Subject 3: ";                // Prompt for subject 3
cin >> g3 >> ch3;                     // Input grade and credit hours
double gp1 = getGradePoint(g1);      // Convert grade to points
double gp2 = getGradePoint(g2);      // Convert grade to points
double gp3 = getGradePoint(g3);      // Convert grade to points

double gpa = (gp1 * ch1 + gp2 * ch2 + gp3 * ch3) / (ch1 + ch2 + ch3); // GPA formula
cout << "Semester GPA: " << gpa << endl; // Display GPA

return 0;                             // End of program
}
```